

1) Image Enhancement in Spatial Domain

→ Image enhancement improves quality of an image by modifying the visual pixel values.

→ Gray-level transformation adjusts brightness and contrast.

→ Histogram processing improves image contrast.

→ Spatial filtering uses smoothing filters to remove noise and sharpening filters to enhance edges.

2) Image Smoothing vs Image Sharpening

→ Image smoothing reduces noise and blur using low-pass filters such as Mean and Gaussian filters.

→ Image sharpening enhances edges and fine details using high-pass filters such as Sobel and Laplacian.

Smoothing is used for noisy images,
sharpening is used to improve
object clarity.

3) Image Enhancement In Frequency Domain.

Frequency domain enhancement uses the
Fourier Transform to process images.

Low-pass filters remove high-frequency noise and smooth the image.

High-pass filters enhance edges and fine details by preserving high-frequency components.

Used in image restoration, noise reduction and image enhancement.

1) Image Segmentation.

→ It Divides an Image into meaningful regions for analysis.

• Techniques include point detection, line detection, edge detection, thresholding and region-based segmentation.

• Threshold is simple and fast, while region-based segmentation gives more accurate results.

• Applications include medical imaging, object recognition and also the Driver Detection.

2) Edge Detection, Edge Linking and Boundary Detection.

→ Edge Detection identifies intensity changes to find the object edges.

→ Edge linking connects broken edge pixels to form the continuous edges.

→ Boundary Detection extracts the complete outline of objects.

→ These techniques improve object recognition, image segmentation, face recognition, and industrial defect detection.